Electronic Space Division Switching

- Stored Program Control
- Centralized SPC
- Distributed SPC
- Software Architecture
- Application Software
- Enhanced Services
- Two-Stage Networks
- Three-Stage Networks
- N-Stage Networks

Electronic Space Division Switching

- Efforts to improve the speed of control and signaling inLate 1940s and early 1950s.
- Use of vacuum tubes, transistors, gas diodes, magnetic drums and cathode ray tubes.
- Arrival of modern electronic digital computers.
- The registers and translators of common control systems can be replaced by a single digital computer.

Stored Program Control

Stored Program Control (SPC)

- Carrying out the exchange control functions through programs stored in the memory of a computer.

- Consequence
  - ✓ Full-scale automation of exchange functions
  - ✓ Introduction of a variety of new services

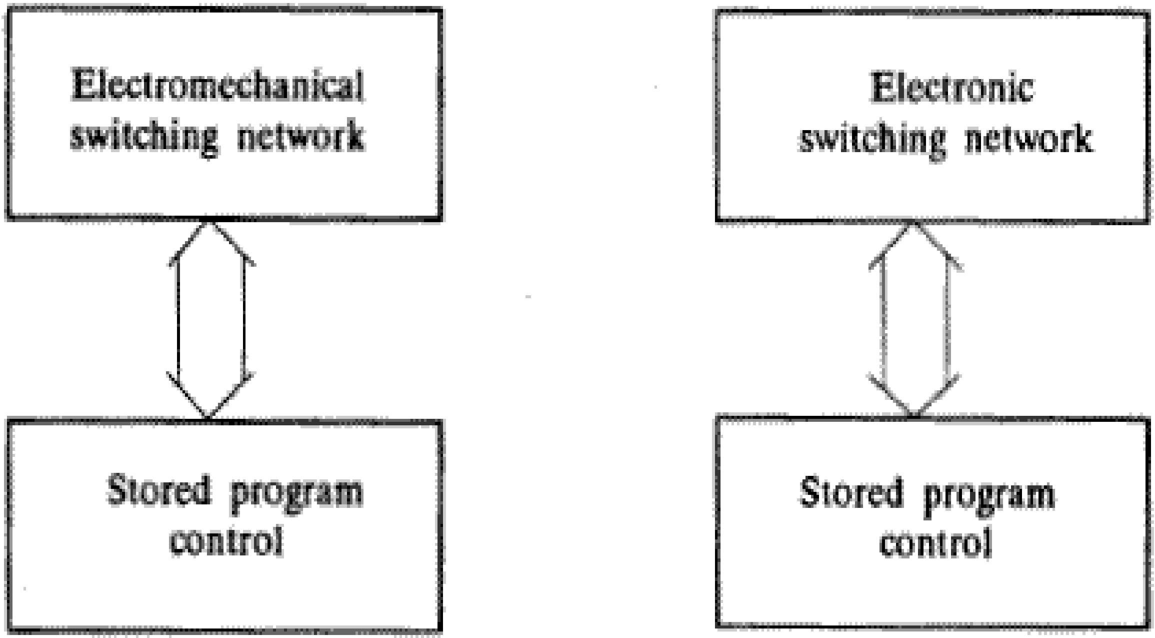
Stored Program Control

- New features possible for SPC
  - ✓Common Channel Signaling (CCS)
  - ✓Centralized maintenance
  - ✓Automatic fault diagnosis
  - ✓Interactive human-machine interface
- Special Requirements of SPC
  - ✓ Operating without interruption
  - ✓ Fault tolerant hardware and software

Stored Program Control

- Two types of SPC switching system
  - ✓ Electromechanical Switching SPC+Electromechanical switching network
  - ✓ Electronic Switching SPC+Electronic switching network

Stored Program Control



(a) Electromechanical switching (b) Electronic switching

Stored Program Control

- Organization of SPC
  - ✓ Centralized SPC Broadly used in early SPC switching systems.
  - ✓ Distributed SPC Gaining popularity in modern switching systems.

Centralized SPC

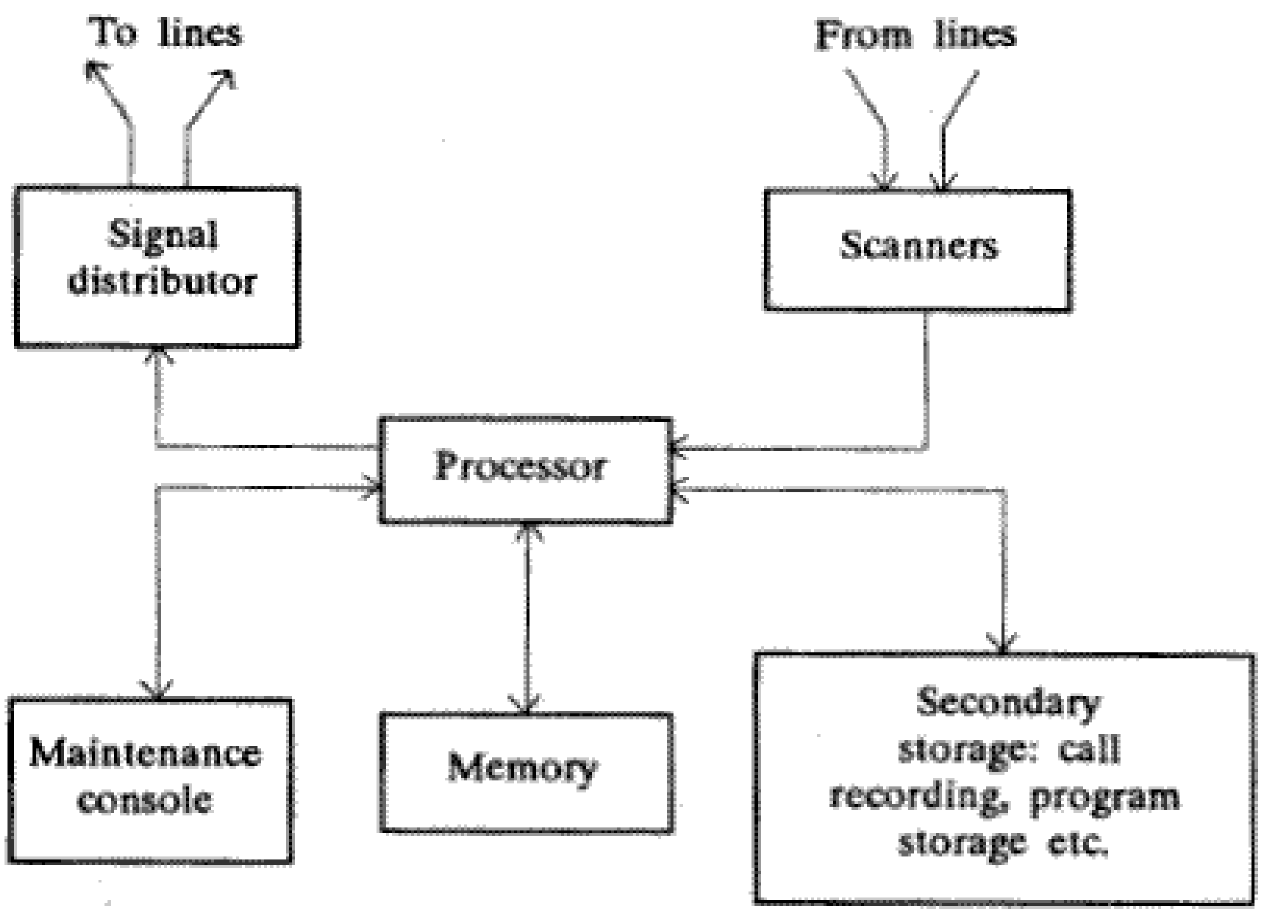
➢ Concept

- All the control equipment is replaced by a single powerful processor.

➢ Configuration of centralized SPC

- Typical organization
- Redundant configuration

Centralized SPC



Distributed SPC

- Concept of distributed SPC
  - The control functions are shared by many processors within the exchanges.
- Background
  - Low cost processors
- Advantages
  - Better Availability
  - Better Reliability
- Decomposition of Control Functions
  - Vertical decomposition
  - Horizontal decomposition

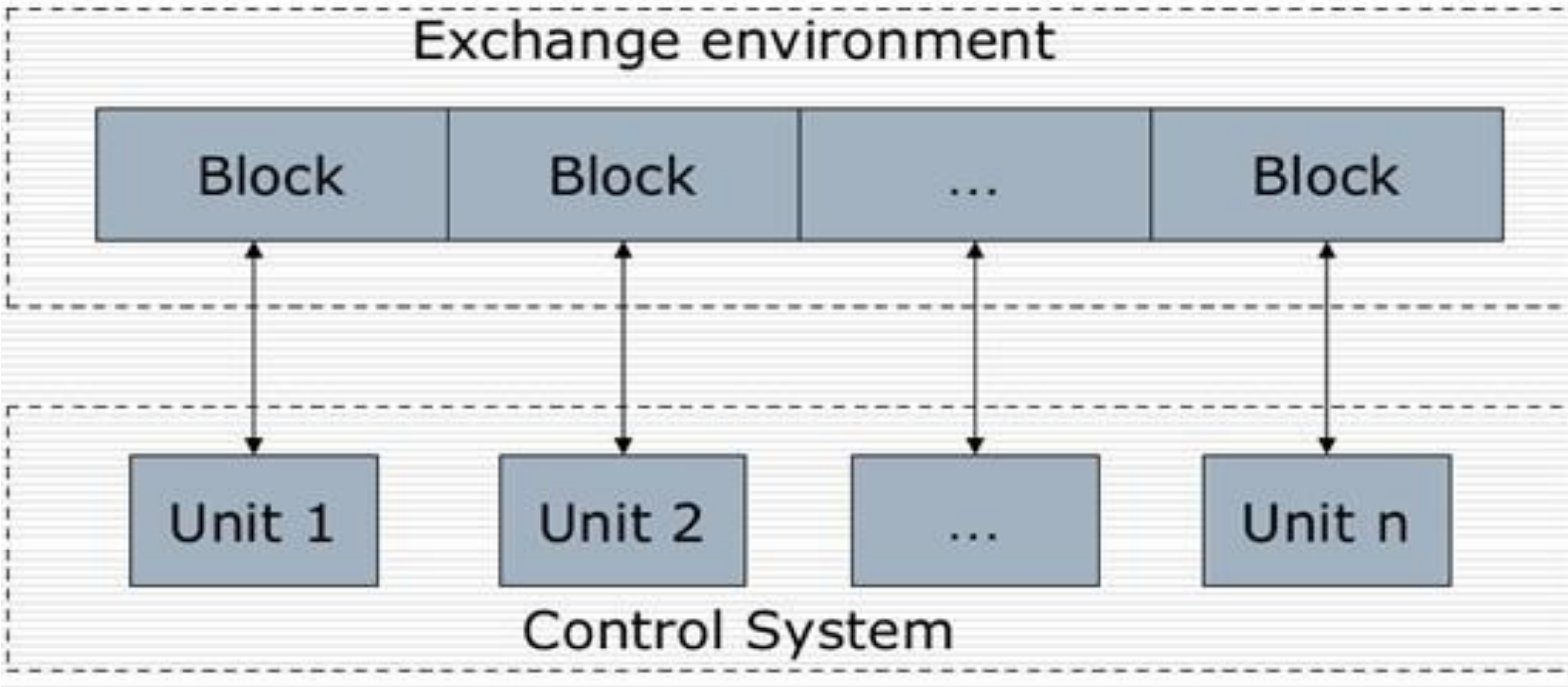
Distributed SPC

❑Vertical decomposition

The exchange environment is divided into several blocks

- ✓ Each block is assigned to a processor.
- ✓ A processor performs all control functions related to the corresponding block.
- ✓ The processor in each block may be duplicated for redundancy purposes.
- ✓Obviously, the control system consists of a number of control units.
- ✓ The modular structure is flexible for system expanding.

Distributed SPC

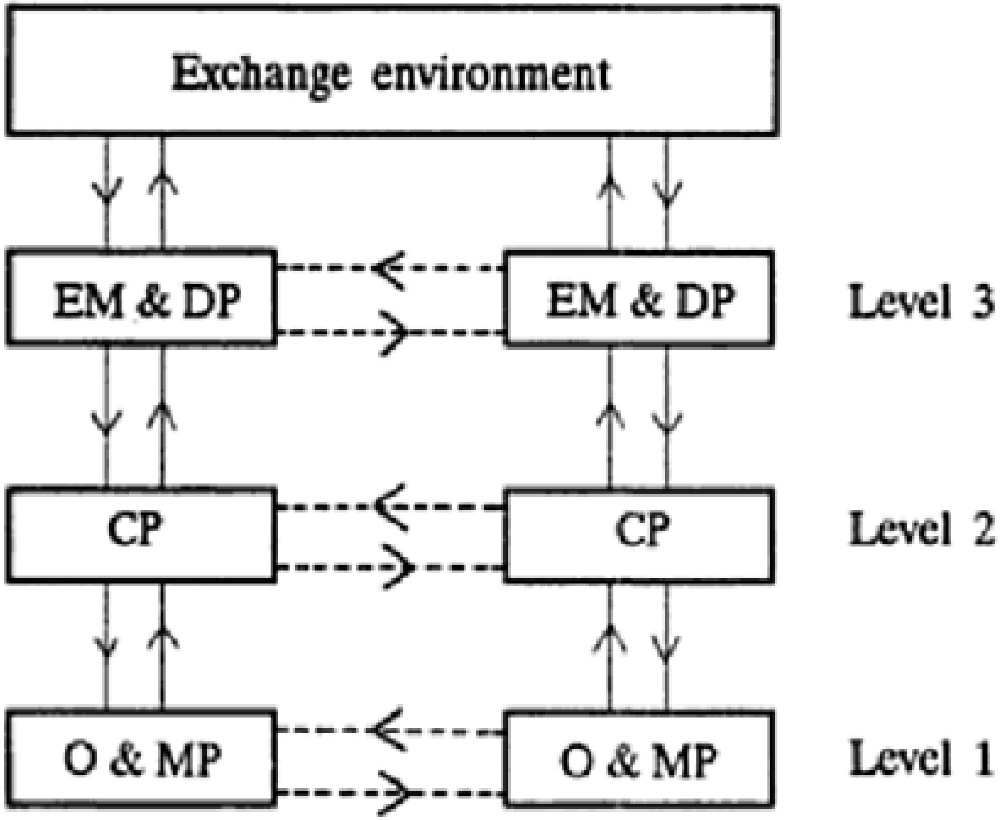


Distributed SPC

❑Horizontal decomposition

- ✓The control functions are divided into groups, e.g. event monitoring, call processing, and O&M functions.
- ✓ Each processor performs only one or some of the exchange control functions.
- ✓ A chain of processors are used to perform the entire control of the exchange.
- ✓ The entire chain may be duplicated to provide redundancy.

Distributed SPC



CP = call processor EM&DP = event monitoring and distribution processor O&MP = operation and maintenance processor

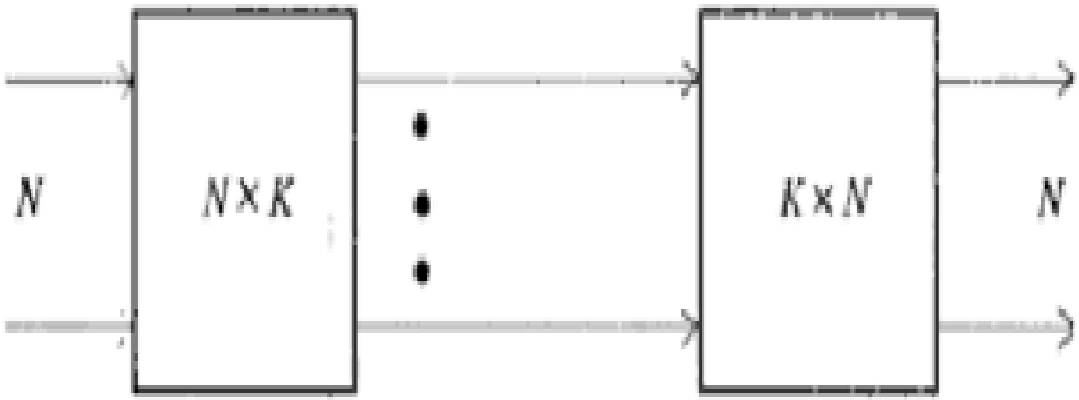
Enhanced Services

- Categories of enhanced services
  - ✓ Services associated with the calling subscriber and designed to reduce the time spent on dialing and the number of dialing errors.
  - ✓ Services associated with the called subscriber and designed to increase the call completion rate.
  - ✓ Services involving more than two parties.
  - ✓ Miscellaneous services.

Two-Stage Networks

- Theorem
  - For any single stage network, there exists an equivalent multistage network.
- Simple Two-stage NxN network
  - A NxN single stage network with a switching capacity of K connections can be realized by a two-stage network of NxK and KxN.

Two-Stage Networks



➢First Stage: Any of the N inlets can be connected to any of the K outputs. NK switching elements.

➢ Second Stage: Any of the K inputs can be connected to any of the N outlets. NK switching elements.

➢ There are K alternative paths for any inlet/outlet pair connection.

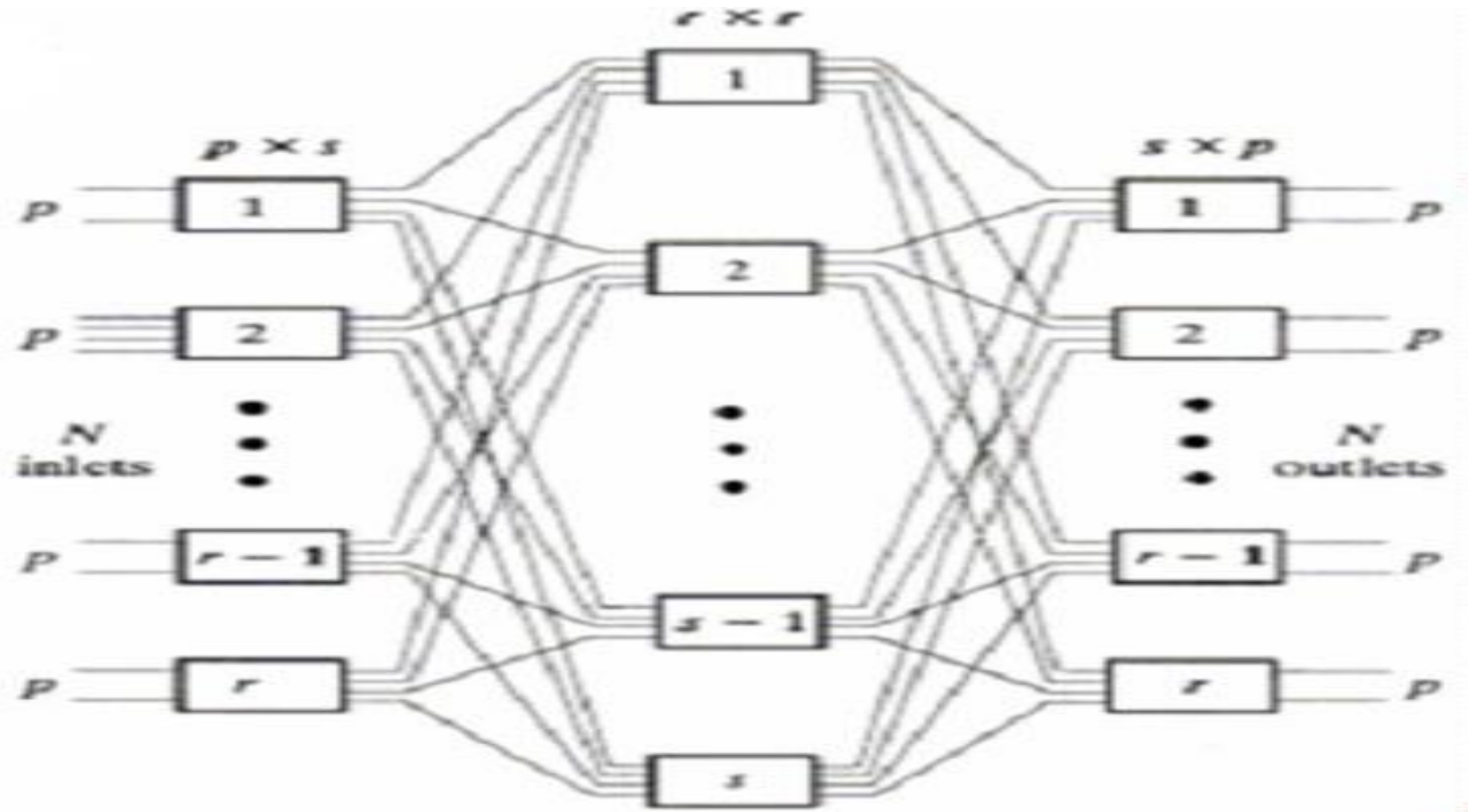
Two-Stage Networks

- ➢ Single stage vs. Multistage networks
  - ✓Inlet to outlet connection
  - ✓Quality of link Utility of cross-points
  - ✓Establishment of a specific connection
  - ✓ Cross-point & path
  - ✓Redundancy Number of cross-points
  - ✓Capacitive loading problem
  - ✓Blocking feature
  - ✓Call establishing time

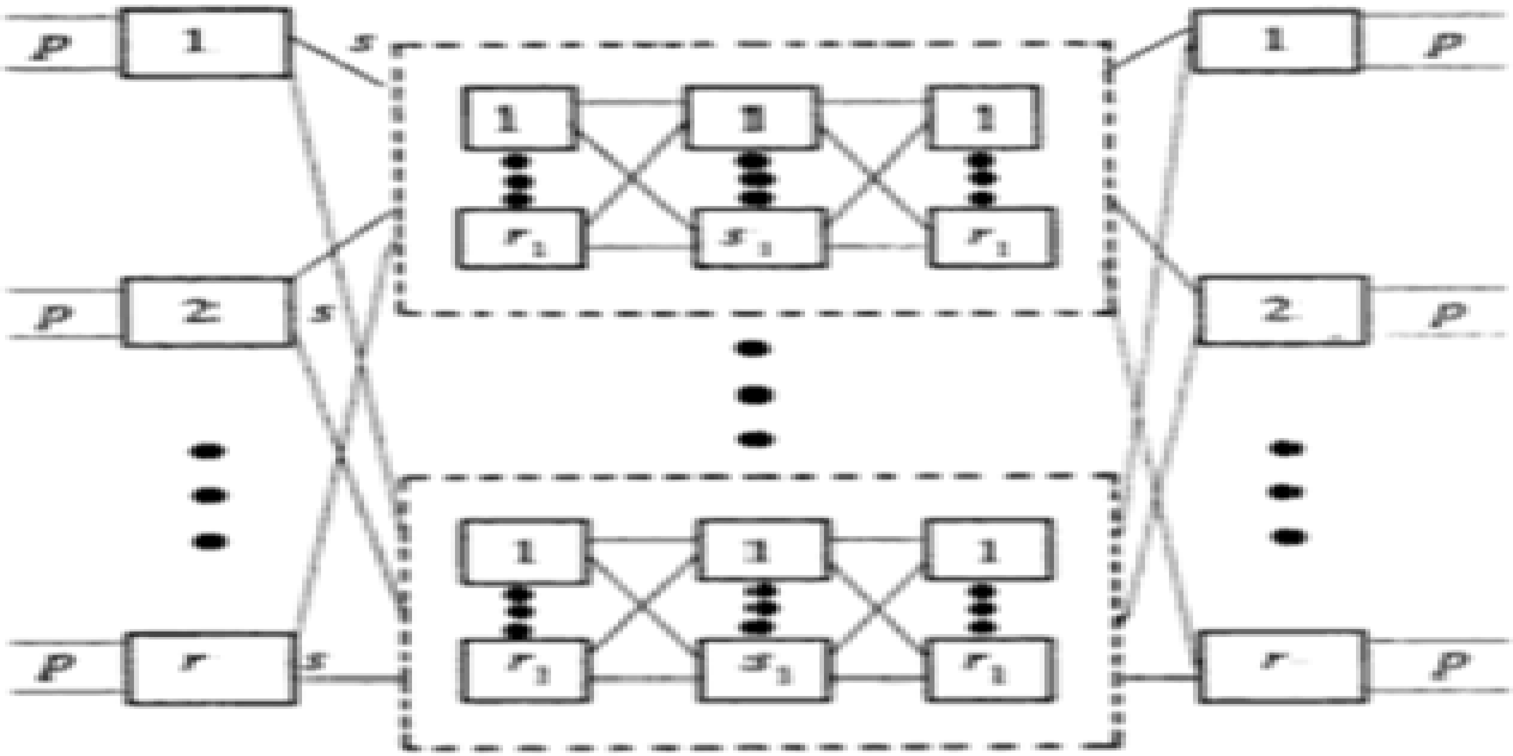
Three-Stage networks

- General structure of an NxN threestage blocking network
  - ✓Stage 1: p x s switching matrices
  - ✓Stage 2: r x r switching matrices
  - ✓Stage 3: s x p switching matrices
  - ✓N=pxr, s is changable
  - ✓Compared with a two-stage network, there are s alternative paths between a pair of inlet and outlet.

Three-Stage networks



n-stage networks



n-stage networks

- Further reduction in the number of switching elements are possible by using even higher number of stages than three.

- Construction of multi-stage networks ? By replacing the middle blocks with three-stage network blocks continually, any number of stages can be obtained.